MEMO

[redacted]

Class Test 4 (Tu)	[redacted]	Time: 20 min	$\overline{}$ 8

Surname: **Initials:**

Student number: **M number:**

Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator.

QUESTION 1 [2]

Evaluate the limit

$$\lim_{x\to 0} \frac{\tan 5x}{7x}.$$

$$\lim_{x\to 0} \frac{\tan 5x}{7x} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{\tan 5x}{5x} \times \frac{5}{7}$$

$$= \frac{5}{7} \lim_{x\to 0} \frac{\tan 5x}{5x} = \frac{5}{7} \times 1 = \frac{5}{7}.$$

QUESTION 2 [2]

Consider the function $f(x) = \sqrt{x}$. Use the definition of the derivative at a point to find $f'(1)$.

$$f'(1) = \lim_{x\to 1} \frac{f(x)-f(1)}{x-1}$$

$$= \lim_{x\to 1} \frac{\sqrt{x} - \sqrt{1}}{x-1} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 1} \frac{\sqrt{x}-1}{x-1} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 1} \frac{\sqrt{x}-1}{(\sqrt{x}-1)(\sqrt{x}+1)} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 1} \frac{1}{\sqrt{x}+1} = \frac{1}{2}$$

QUESTION 3 [2, 2]
Consider the function

$$g(x) = \frac{\sqrt{x^4 + x^2 + 1}}{1 - x^2}. = \frac{\sqrt{x^4+x^2+1}}{(1-x)(1+x)}$$

(a) Find the vertical asymptote(s) of g.

$$\lim_{x \to 1^-} \frac{\sqrt{x^4+x^2+1}}{(1-x)(1+x)} = +\infty$$

$x \to 1^- \Rightarrow x < 1 \Rightarrow x - 1 < 0 \Rightarrow 1 - x > 0$

$\Rightarrow x = 1$ is a vertical asymptote.

$$\lim_{x \to -1^-} \frac{\sqrt{x^4+x^2+1}}{(1-x)(1+x)} \quad \left(\frac{\sqrt{3}}{(2)(-)}\right) = -\infty$$

$x \to -1^- \Rightarrow x < -1 \Rightarrow x + 1 < 0$

$\Rightarrow x = -1$ is a vertical asymptote.

(b) Find the horizontal asymptote(s) of g.

$$g(x) = \frac{\sqrt{x^4+x^2+1}}{1-x^2} = \frac{\sqrt{x^4(1+1/x^2+1/x^4)}}{x^2(1/x^2-1)} = \frac{x^2\sqrt{1+1/x^2+1/x^4}}{x^2(1/x^2-1)} \quad *$$

$$\Rightarrow \lim_{x \to \infty} g(x) = \lim_{x \to \infty} \frac{\sqrt{x^4+x^2+1}}{1-x^2} \quad \left(\frac{\infty}{-\infty}\right)$$

$$= \lim_{x \to \infty} \frac{\sqrt{1+1/x^2+1/x^4}}{\left(\frac{1}{x^2}-1\right)} \quad (\text{from } *)$$

$$= -1$$

$y = -1$ is a horizontal asymptote

$$\lim_{x \to -\infty} g(x) = \lim_{x \to -\infty} \frac{1+1/x^2+1/x^4}{1/x^2-1} \quad (\text{from } *)$$

$$= -1$$

$\Rightarrow y = -1$ is the only horizontal asymptote.